

ML OPS Experimentation Stage

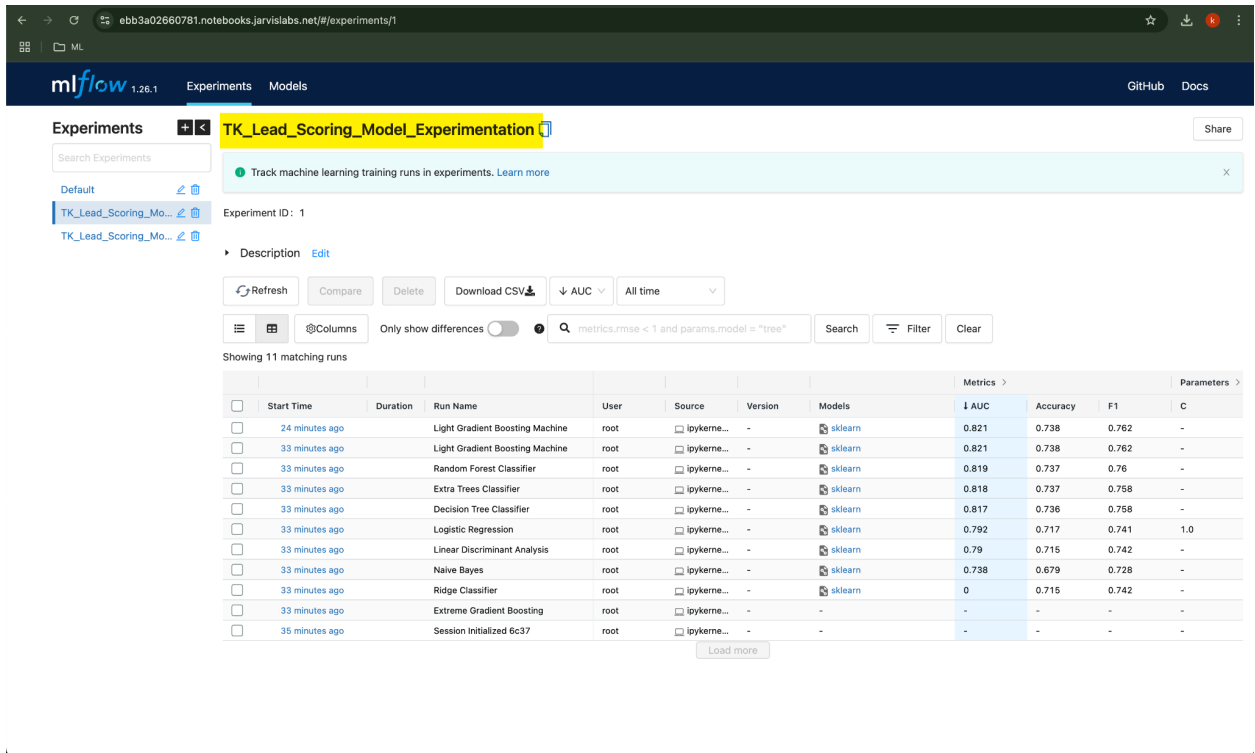
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Running ML Flow Server and Artifacts URI

```
root@ebb3a0e43a2a:~# mlflow server --backend-store-uri='sqlite:///Assignment/02_training_pipeline/notebooks/lead_scoring_model_experimentation.db'
--default-artifact-root='file:///Assignment/02_training_pipeline/notebooks/mlruns' --port=6006 --host=0.0.0.0
[2025-02-13 17:58:03 +0000] [6024] [INFO] Starting gunicorn 20.1.0
[2025-02-13 17:58:03 +0000] [6024] [INFO] Listening at: http://0.0.0.0:6006 (6024)
[2025-02-13 17:58:03 +0000] [6024] [INFO] Using worker: sync
[2025-02-13 17:58:03 +0000] [6026] [INFO] Booting worker with pid: 6026
[2025-02-13 17:58:03 +0000] [6027] [INFO] Booting worker with pid: 6027
[2025-02-13 17:58:03 +0000] [6028] [INFO] Booting worker with pid: 6028
[2025-02-13 17:58:03 +0000] [6029] [INFO] Booting worker with pid: 6029
```

Model -1: TK_Lead_Scoring_Model_Experimentation



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Search Experiments

TK_Lead_Scoring_Model_Experimentation

Experiment ID: 1

Description Edit

Refresh Compare Delete Download CSV AUC All time

Columns Only show differences metrics.rmse < 1 and params.model = "tree" Search Filter Clear

Showing 11 matching runs

	Start Time	Duration	Run Name	User	Source	Version	Models	Metrics >	Parameters >		
								↓ AUC	Accuracy	F1	C
☐	24 minutes ago		Light Gradient Boosting Machine	root	ipykerne...	-	sklearn	0.821	0.738	0.762	-
☐	33 minutes ago		Light Gradient Boosting Machine	root	ipykerne...	-	sklearn	0.821	0.738	0.762	-
☐	33 minutes ago		Random Forest Classifier	root	ipykerne...	-	sklearn	0.819	0.737	0.76	-
☐	33 minutes ago		Extra Trees Classifier	root	ipykerne...	-	sklearn	0.818	0.737	0.758	-
☐	33 minutes ago		Decision Tree Classifier	root	ipykerne...	-	sklearn	0.817	0.736	0.758	-
☐	33 minutes ago		Logistic Regression	root	ipykerne...	-	sklearn	0.792	0.717	0.741	1.0
☐	33 minutes ago		Linear Discriminant Analysis	root	ipykerne...	-	sklearn	0.79	0.715	0.742	-
☐	33 minutes ago		Naive Bayes	root	ipykerne...	-	sklearn	0.738	0.679	0.728	-
☐	33 minutes ago		Ridge Classifier	root	ipykerne...	-	sklearn	0	0.715	0.742	-
☐	33 minutes ago		Extreme Gradient Boosting	root	ipykerne...	-	-	-	-	-	-
☐	35 minutes ago		Session Initialized 6c37	root	ipykerne...	-	-	-	-	-	-

Load more

ebb3a02660781.notebooks.jarvislabs.net/#/experiments/1/runs/edb384dbf24b4821a951c8d5c8369807

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TK_Lead_Scoring_Model_Experimentation > Light Gradient Boosting Machine

Light Gradient Boosting Machine

Date: 2025-02-13 23:40:00 Source: `ipykernel_launcher.py` User: root

Status: UNFINISHED Lifecycle Stage: active Parent Run: 562aaa73fa564c9aaebaf497ace6457c

► Description [Edit](#)

► Parameters (20)

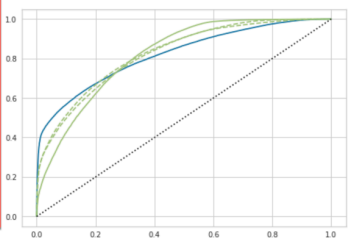
► Metrics (8)

► Tags (5)

▼ Artifacts

- model
 - MLmodel
 - conda.yaml
 - model.pkl
 - python_env.yaml
 - requirements.txt
 - AUC.png
 - Confusion Matrix.png
 - Feature Importance.png
 - Holdout.html

Full Path: file:///Assignment/02_training_pipeline/notebooks/mruns/1/edb384dbf24b4821a951c8d5c8369807/artifacts/AUC.png
Size: 27.41KB



Model - 2: TK_Lead_Scoring_Model_Experimentation_After_Dropping_Features

ebb3a02660781.notebooks.jarvislabs.net/#/experiments/2

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Experiments

Search Experiments

Default TK_Lead_Scoring_Mo... TK_Lead_Scoring_Mo...

TK_Lead_Scoring_Model_Experimentation_After_Dropping_Features

Experiment ID: 2

Description Edit

Refresh Compare Delete Download CSV Start Time All time

Only show differences metrics.rmse < 1 and params.model = "tree" Search Filter Clear

Showing 12 matching runs

	Start Time	Duration	Run Name	User	Source	Version	Models	AUC	Accuracy	F1	C
	39 minutes ago		Session Initialized 92bc	root	ipykerne...	-	-	-	-	-	-
	33 minutes ago		Light Gradient Boosting Machine	root	ipykerne...	-	sklearn	0.821	0.739	0.76	-
	35 minutes ago		Light Gradient Boosting Machine	root	ipykerne...	-	sklearn	0.821	0.739	0.762	-
	36 minutes ago		Naive Bayes	root	ipykerne...	-	sklearn	0.734	0.673	0.725	-
	36 minutes ago		Linear Discriminant Analysis	root	ipykerne...	-	sklearn	0.773	0.7	0.727	-
	36 minutes ago		Ridge Classifier	root	ipykerne...	-	sklearn	0	0.7	0.727	-
	36 minutes ago		Logistic Regression	root	ipykerne...	-	sklearn	0.784	0.71	0.74	1.0
	36 minutes ago		Decision Tree Classifier	root	ipykerne...	-	sklearn	0.817	0.736	0.758	-
	36 minutes ago		Extra Trees Classifier	root	ipykerne...	-	sklearn	0.818	0.736	0.758	-
	36 minutes ago		Random Forest Classifier	root	ipykerne...	-	sklearn	0.819	0.737	0.759	-
	36 minutes ago		Extreme Gradient Boosting	root	ipykerne...	-	sklearn	0.821	0.738	0.761	-
	36 minutes ago		Light Gradient Boosting Machine	root	ipykerne...	-	sklearn	0.821	0.739	0.762	-

Load more

ebb3a02660781.notebooks.jarvislabs.net/#/experiments/2/runs/75761c876d0548c7be88d2bcd8d06e52

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TK_Lead_Scoring_Model_Experimentation_After_Dropping_Features > Light Gradient Boosting Machine

Light Gradient Boosting Machine

Date: 2025-02-13 23:46:05 Source: ipykernel_launcher.py User: root

Status: UNFINISHED Lifecycle Stage: active Parent Run: 658a893e1e7041deb47ed87daf3c989c

Description Edit

Parameters (23)

Metrics (8)

Tags (5)

Artifacts

model

- MLmodel
- conda.yaml
- model.pkl
- python_env.yaml
- requirements.txt
- AUC.png
- Confusion Matrix.png
- Feature Importance.png
- Holdout.html
- Iterations.html

Full Path: file:///Assignment/O2_training_pipeline/notebooks/mlruns/2/75761c876d0548c7be88d2bcd8d06e52/artifacts/model

Register Model

MLflow Model

The code snippets below demonstrate how to make predictions using the logged model. You can also register it to the model registry to version control

Model schema

Input and output schema for your model. Learn more

Name	Type
No schema. See MLflow docs for how to include input and output schema with your model.	

Make Predictions

Predict on a Spark DataFrame:

```
import mlflow
logged_model = 'runs:/75761c876d0548c7be88d2bcd8d06e52/model'

# Load model as a Spark UDF. Override result_type if the model does not return double values.
loaded_model = mlflow.pyfunc.spark_udf(spark, model_uri=logged_model, result_type='double')

# Predict on a Spark DataFrame.
columns = list(df.columns)
df.withColumn('predictions', loaded_model(*columns)).collect()
```